Explainable AI for Credit Risk Scoring in Digital Lending: A Comparative Analysis of Interpretability Frameworks

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***Abstract*—The assessment of credit risk has been revolutionized by the integration of Artificial Intelligence(AI) and Machine Learning(ML) in the financial institutions, with advancement in the high predictive precision. However, due to the "Black Box" nature of the high-performance models such as XGBoost and LightGBM, there have been certain significant challenges regarding transparency and regulatory compliance and in addition to ethical fairness as well. This research addresses the critical gap between predictive power and interpretability by proposing a comprehensive Explainable AI (XAI) framework. By integrating SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), ergo the study evaluates the "faithfulness" (trustworthiness) and stability of model decisions in a digital lending environment. The research study utilizes the credit risk dataset from Kaggle, demonstates a hybrid feature engineering approach on the dataset and preserves accuracy while provding "reason codes" in laymen terms for loan approvals and rejections. The aim of the findings is to establish a standardardized evaluation of XAI tools, ensuring that advanced credit scoring remains fair, accessible and accountable to underserved populations.**

***Index Terms*—Explainable AI, Credit Risk Scoring, XAI, SHAP, LIME, Digital Lending**

I. Introduction

A digital metamorphosis is what the world of finance is currently experiencing, facilitated by the usage of automated decision-making processes. In today's banking era [1], the process of analzying credit risks is not simply manual review of accounting books anymore rather it is a fast-paced computation process using large automated models. Given that financial inclusivity [2] is a critical component of modern economic development, the capacity for assessing one's creditworthiness plays an essential role.

To achieve the same level of precision [3], financial institutions need to adopt more advanced machine learning models compared to simply Logistic Regression, which brings the spotlight to the models discussed in this research study; XGBoost and LightGBM [4]. They are Neural Network Architectures that are capable of analyzing a vast array of non-linear features. But, this brings us to the complexity of the situation that as the models grow in complexity they become less understable to users which results in the concept of "Black Box", in which the decision process cannot be explained using mathematics alone.

Currently, all solutions available to this "Black Box" model problem involves the usage of post-hoc explainer models. The two models included are SHAP (SHapley Additive exPlanations), which is based on concepts from game theory, and LIME (Local Interpretable Model-agnostic Explanations). Both SHAP and LIME are commonly used in practice to explore the hidden layer of the black box, basically to find out what is going on inside it [5]. However, these techniques are often treated as add-ons that are applied after model training instead of using it as a core component of model development lifecycle.

Most researchers are determined to make the AI "smart" using scores such as AUC-ROC and F1-score to measure its precision and accuracy. But, they treat "the explainability" [6] part as an afterthought. It is like building a high-speed car but forgetting to put a dashboard that would tell the driver whether they are going fast or if the engine is overheating. There is no logical explanation behind the decisions made by the black box models.Therefore, if two different tools give two different reasons for a loan rejection, we don't have a standardized way to figure out which one is right. No-one has yet created a perfect system that is both super-accurate and easy to explain in a way that is mandated by the regulators and from an ethical and moral point of view.

Notwithstanding the widespread use of these methodologies [7], there are still numerous research gaps in this field. Firstly, SHAP and LIME do not always agree with each other on the explanation for the same data point, leaving banks without "ground truth." Secondly, there is no clear "rulebook" regarding the quality metrics for good explanations. However, most importantly, XAI outputs generated by these methodologies are represented in the form of complex mathematical vectors and codes incomprehensible even to layman data scientists.

The paper conducts an evaluation of SHAP and LIME methods by ensuring that they boost algorithms and explains the criteron of two parameters, namely "Faithfulness" to the model and "Interpretability" for the common person. This paper ensures that the "reason codes"(explanation) provided

by Artificial Intelligence will have legal and ethical legitimacy.

The rest of this paper is arranged as follows: Section II provides the background information on studies and papers about algorithmic bias and XAI practices in the past. Section III elaborates on the method adopted in the experiment, detailing information about the processes of data preparation and formation of a hybrid feature engineering strategy. Section IV elaborates the experimental procedures, demonstrating the setup and implementation of the XAI methods. In section V, comparative analysis is conducted on the results in the domain of fair lending.

A. Main Contributions and Novelty

The main contributions of this research paper are as follows:

- **Propose** a comprehensive Explainable Artificial Intelligence framework to cater and bridge the gap between high-performance ”black-box” models and regulatory transparency, which is specifically optimized for digital lending.
- **Design** a hybrid feature engineering method that utilizes credit risk knowledge alongside automated selection to maintain high predictive accuracy while ensuring the feature of interpretability.
- **Analyze**the trade-off between model ”faitfulness” and stability, basically providing a comparative analysis of interpretability frameworks on a standardized credit risk dataset.
- **Develop** a ”Reason Code” generation mechanism that explains complex vector mathematical vectors into laymen terms or to put into better words; human-readable justifications for loan approvals and rejections.
- **Evaluate** the impact of Explainable AI integration on financial inclustivity by demonstrating how transparent decision-making can mitigate algorithmic bias against underserved populations.

II. Problem Statement

In traditional banking, when a borrower used to apply for a loan the bank would use statistical methods such as Logistic Regression or FICO score in order to analyze their credit worthiness. But now because banks use AI and ML, these models look at thousands of tiny details in order to predict if the loan will be repaid by the borrower or not, basically checking their credit worthiness. These models are very complex, hence they are called “Black Boxes”. Even developers cannot explain why they rejected a specific person. Think of it as a robot that made a decision but stays silent about its reasons. There are certain statistics which have shown that algorithmic bias can have real-world consequences. The affected industries include peer-to-peer lending platforms, microfinance institutions, and digital payment companies. The societal impact is vast as millions of creditworthy individuals, particularly in undeserved and developing communities, are denied access to financial services due to non-transparent decisions made by an algorithm that they cannot challenge, understand, or hold accountable. The problem therefore represents an outmost pressing research challenge which sits at the intersection of technology, regulation, fairness, and financial inclusion.

III. Literature Review

Most researchers are using the “After-the-Fact” Explaners [8] approach where the “Black Box” model is analyzed to try and guess why it did what it did. The tools used for this approach are called SHAP and LIME.

A model called XGBOOST [9] was found in order to build a middle ground. Some researchers claimed that; “Why build a Black Box at all? Let’s build models that are clear from the start.” Because of that, XGBOOST was built. They discovered that certain info like gender and age can be removed from the AI’s memory and the AI stays just as smart. As a consequence, this proves that there is no need to be “unfair” in order to be accurate.

Researchers linked AI design to real laws (like the GDPR in Europe). They argued that there was a need for “Explainability Audits” to make sure banks aren’t cheating or frauding. A study showed that if the AI was forced to be fair from the start, it helps more people to get loans who were previously ignored.

IV. Proposed Methodology

A. Data Acquisition and Preprocessing

In this particular stage, integration of multiple financial data sources will be done which will include traditional credit information and digital lending data. This step is very important because it provides a clear and concise understanding of a borrower’s credit profile. Moreover, preprocessing tasks include the management of missing values, scaling numerical values, and since only numerical values will be dealt with there will be no mention of categorical data. This phase is major because the effectiveness of the XAI framework depends heavily on the quality and reliability of the dataset that will be fed to it. It also prepares a structured and clean dataset for the engineering stage which will be later on used by decision-makers.

B. Hybrid Feature Engineering

This component will ensure highly predictive features while making sure that they are interpretable and explainable. A hybrid strategy will be used by combining statistical techniques such as SHAP and LIME with credit domain knowledge. Feature engineering helps in reducing dimensionality found in models and eliminates noise which if not done would result in unclear or fragmented explanations in XAI models. By the conversion of raw data into meaningful indicators, this stage helps ML models to capture patterns and sequences in the decision process that can be understood later on.

C. Black-Box Model Development

XGBoost and LightGBM are trained to estimate the likelihood of default at this phase by now. These models are selected because of their strong predictive capabilities especially for complex and non-linear financial data thereby outperforming traditional statistical methods such as logistic regression. They act as the core computational engine of the system. They provide

TABLE I
PROPOSED SOLUTIONS IN LITERATURE — EXPLAINABLE AI FOR CREDIT RISK SCORING IN DIGITAL LENDING

Model	Experiment	Problem	Contribution	Limitation
XGBoost + SHAP/LIME [10]	Real loan data; regulator reports	No reason given for loan denial	Plain-language decision reports	Tested on Western data only
Random Forest + SHAP [11]	9 models on Credit Risk	Unknown AI factors in default prediction	Age and job length are top predictors	Single dataset; limited generalization
LightGBM + SHAP/LIME [12]	14 years of loan data; 87% accuracy	Imbalanced data hurts fairness	Balancing improves accuracy and explainability	SHAP and LIME conflicted in some cases
SHAP + LIME on P2P models [13]	LendingClub; local and global explanations	P2P lending AI unexplained	First XAI study on P2P loan decisions	Legal compliance not evaluated
Surrogate decision tree [14]	Turkish bank data; 91% match rate	Neural networks unauditable	Simple copy explains complex model	Accuracy drops for highly complex models
Multi-model comparison [15]	Credit card default dataset	No accuracy-vs-explainability benchmark	XGBoost best balances both	No XAI tool built or tested
Map of how AI discriminates in credit scoring [16]	Reviewed studies linking bias types to laws like GDPR and ECOA	No clear map existed connecting AI bias to legal rules	Shows exactly which laws apply when AI makes a biased credit decision	Only analyzed laws; did not build or test any technical solution
Early test of deep learning vs. traditional AI for loan default [17]	Compared logistic regression, SVM, and neural networks on LendingClub	Deep learning was never properly benchmarked for credit scoring	Set the first benchmark showing deep learning works for credit risk	Published before modern explanation tools existed; partly outdated
Bias-to-law mapping [18]	Studies linked to GDPR and ECOA	No map of AI bias to legal rules	Shows which laws apply to which bias types	No technical solution built or tested
Hybrid AI model that is both accurate and fair [19]	Removed gender and age from the model; re-tested accuracy	Removing sensitive data like gender was thought to hurt accuracy	Proved that removing sensitive info does not reduce accuracy at all	Only tested on one P2P dataset; needs broader testing
Bias mitigation review [20]	Three mitigation methods compared	No single method works for all groups	Catalogs all major bias-fixing methods	Fixing one bias often breaks another

high accuracy which is much needed although they are difficult to interpret.

The prediction of credit risk is formulated as a probabilistic classification problem. The model estimates the likelihood of default based on input features, and the final decision is obtained as:

$$y^* = \arg\max_{y \in \{0,1\}} P(y \mid X) \tag{1}$$

where X represents the borrower feature vector and $P(y \mid X)$ denotes the predicted probability of class y. The model assigns the label corresponding to the highest probability, indicating either default or non-default.

D. XAI Integration (SHAP vs. LIME)

This is the phase where post-hoc explainability methods such as SHAP and LIME are applied to the XAI models. The goal is to convert complex model outputs with higher accuracy into understandable insights and decisions at both the local and global levels. SHAP ensures consistency using game-theoretic principles, while LIME generates quicker, locally interpretable approximations. This stage connects model predictions with user understanding by producing feature importance insights and decision explanations.

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i), \quad f_k \in \mathcal{F} \tag{2}$$

where $\hat{y}_i$ represents the predicted output for the i-th instance, and x_i denotes the corresponding input feature vector. Each function f_k corresponds to an individual decision tree in the ensemble, and $\mathcal{F}$ represents the space of all possible regression trees. The final prediction is obtained by aggregating the outputs of K such trees, enabling the model to capture complex non-linear relationships in the data.

E. Comparative Evaluation and Dashboarding

This is the final stage which will involve comparing the explainability methods based on evaluation metrics and criteria such as faithfulness (accuracy), stability and ease of understanding. A decision-support system is developed to provide not only credit scores but also clear "reason codes" for approvals or rejections. This is highly important because this step will ensure that the XAI framework makes decisions that comply with the legal requirements like GDPR. This component connects technical modeling of the XAI with practical application by creating a dashboard that enables credit analysts to review and validate model decisions before finalizing loans.

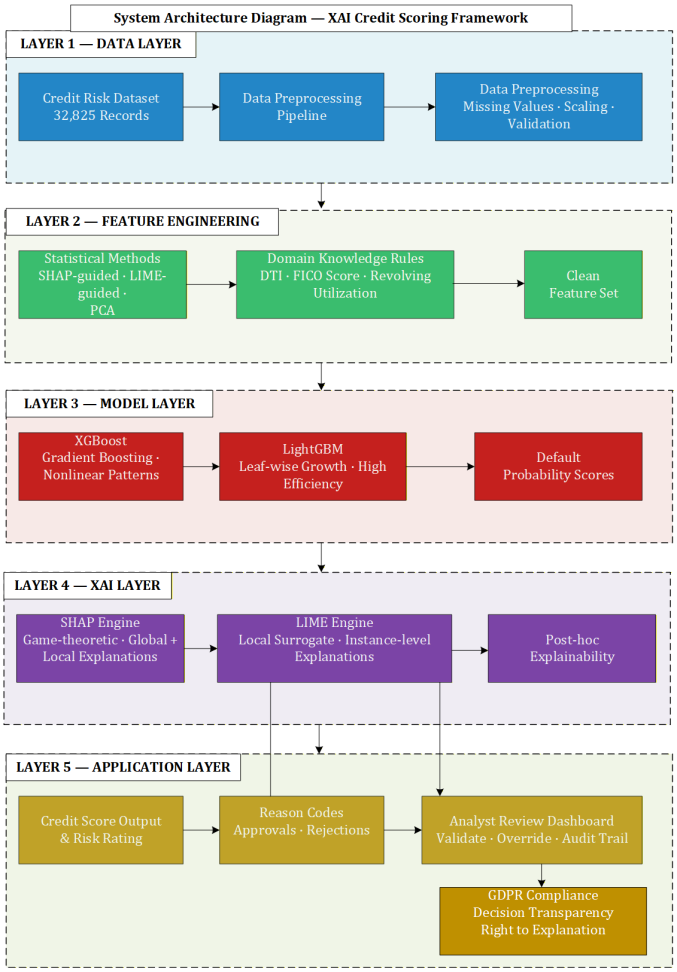

Fig. 1. System Architecture

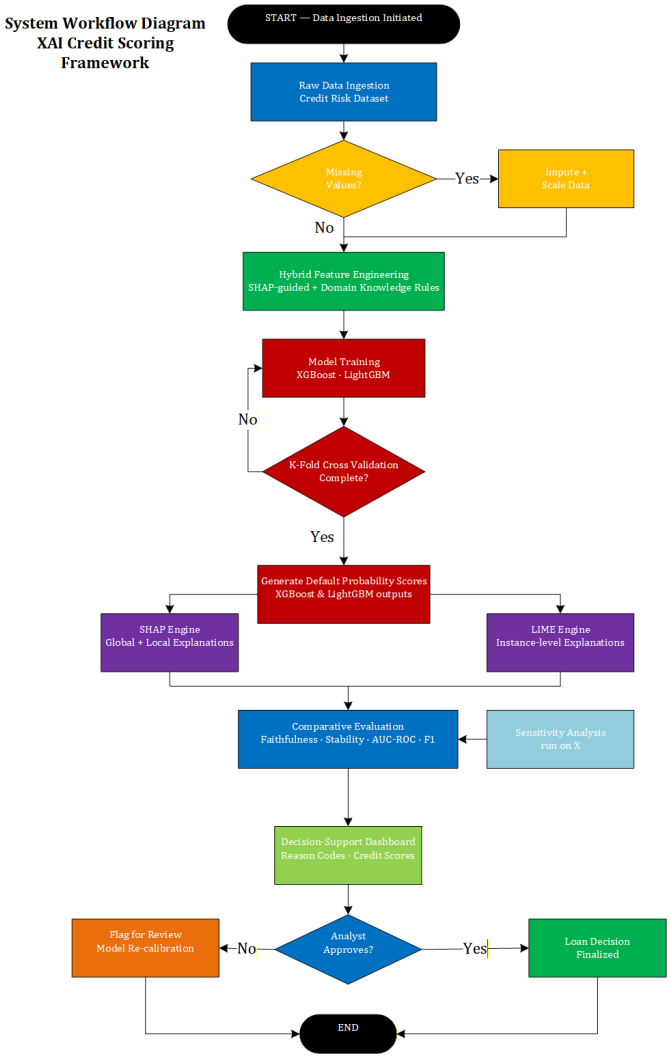

Fig. 2. System Workflow

V. Implementation Details

The proposed system was implemented using Python in Google Colab. Since, Google Colab is a free cloud-based Jupyter Notebook that excluded the hardware limitation. The dataset that will be computed in the study was large, thus Google Colab was used. Various data tools such as Pandas and NumPy were used for data manipulation and numerical computation in addition to data cleaning and handling missing values which is one of the common problem found in majorly large datasets. Powerful decision-tree models; XGBoost and LightGBM help in navigating in structured financial data. They were chosen because both show strong performance on tabular data and handle large datasets well. Why both models? In order to compare performance and evaluate trade-offs (their speed and accuracy). For model development, SHAP and LIME were used to explain why the Artificial Intelligence decided to approve or reject the loan rather than it being just a ”black box”. Basically, interpreting the explainability of the credit risk model.

SHAP explains interpretability at a global and at a local level and on the other hand, LIME explains individual predictions. They both appear to suggest transparency and trust which is higly valued in finance.

Credit Risk Dataset:

The dataset contains 32,582 records. Before the system can read the dataset, missing values will be handled, categorical variables will be encoded and feature scaling will be done. Additionally, the dataset mimics real-world bureau data.It includes features such as person_income, loan_intent, loan_grade, loan_percent_income, loan_amount, percent_homeownership, person_emp_length, loan_int_rate etc.

VI. Results and Discussion

The data set was divided into 80% and 20% which would be used by the model to study patterns and testing, respectively.

XGBoost and LightGBM models were trained with default hyperparameters because they tend to show fair comparison. The experiment involved training the models, making predictions on the test data, and comparing their performance using evaluation metrics.

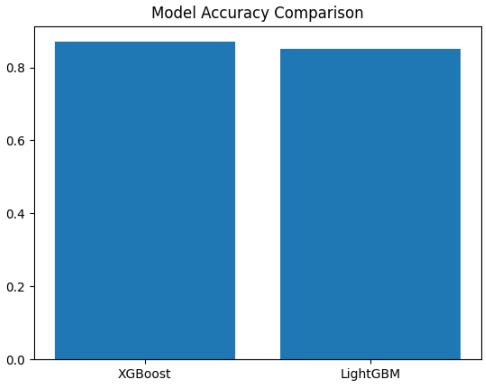

Fig. 3. Model Accuracy Comparison

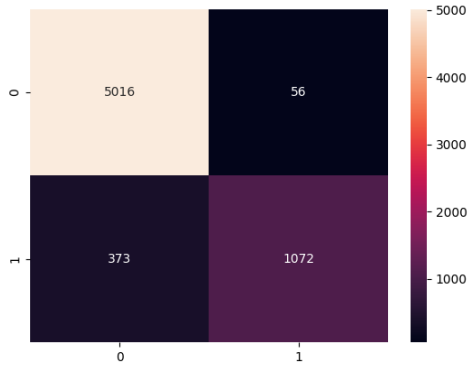

Fig. 4. Confusion Matrix

A. Evaluation Metrics

The performance of the models was evaluated using standard classification metrics as stated below:

Accuracy: Measures the overall correctness of the Model. For determination of the proposed system, accuracy alone is not enough.

Precision: Measures the proportion of correct positive predictions. As precision will ensure that approved loans are less likely to default.

Recall: Measures the ability of the model to identify positive instances. Recall is very important in credit risk prediction because it determines whether the model can identify high-risk borrowers or not.

F1 Score: It shows the harmonic mean of precision and recall. Basically a balanced score of precision and recall.

These metrics were chosen because they provide a comprehensive evaluation of classification performance, especially for credit risk prediction tasks.

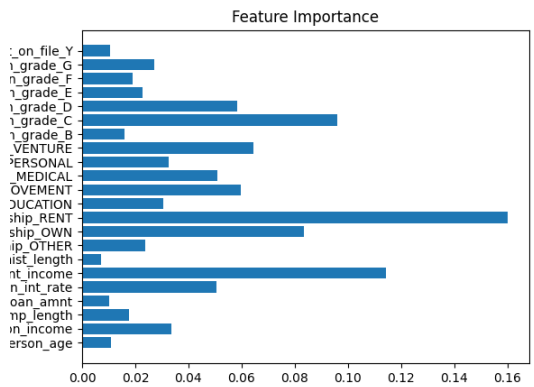

Fig. 6. Feature Importance

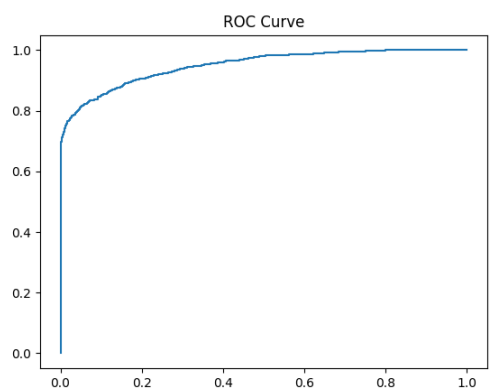

Fig. 5. ROC Curve

B. Model Performance Table

TABLE II
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1 Score
XGBoost	0.93	0.95	0.74	0.83
LightGBM	0.93	0.96	0.73	0.83

C. Results Analysis

The results demonstrate that XGBost and LightGBM exhibit identical overall accuracy and F1 scores. But XGBoost provides a better balance for imbalanced data, offering higher recall; whereas LightGBM precision is sightly better than XGBoost. Therefore, the final verdict is that XGBoost is best for sensitivity,

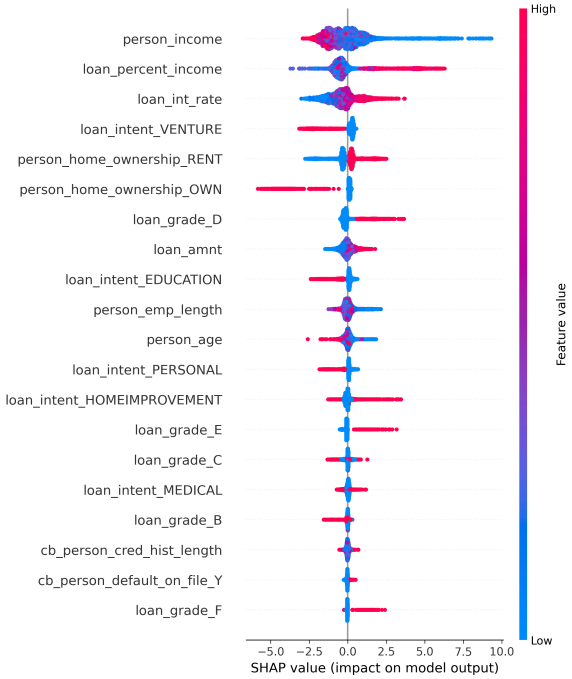

Fig. 7. SHAP Analysis

basically at finding positive instances, and LightGBm is best for precision as it is slightly better at finding negative instances.

In fig. 4. , the confusion matrix appears to show that the model correctly classifies the majority of instances, with fewer misclassifications.And in fig. 5. , the ROC curve suggests a strong discriminative ability.

In fig.6. , feature importance analysis seem to reveal that the financial attributes such as person_income, loan_intent, loan_grade,loan_percent_income, and loan_amount significantly influence predictions. SHAP analysis appears to further enhance interpretability by showing how individual features contribute to each prediction.

VII. COMPLEXITY AND SCALABILITY ANALYSIS

Although both algorithms are efficient when used for tabular data sets, LightGBM exhibited far more superior training performance and results due to its leaf-wife growth algorithm, making it highly efficient and better suited for large-scale data processing tasks. Nevertheless, the use of SHAP comes with considerable extra cost ($O(2^n)$ for precise computations), which means that to be able to provide instant accurate credit decisions in real time, approximation techniques need to be used.

VIII. Limitations

One of the limitations of the proposed system is that the dataset contains missing or imbalanced values which are affecting the results. Moreover, the models seem to be using default hyperparameters which appear to be not yielding optimal performance. Additionally, the computational complexity is another problem when using advanced models and explainability techniques such as SHAP.

IX. Conclusion

This research addressed the critical challenge of the ”black box” nature of AI in financial services by evaluating Explainable AI (XAI) frameworks within credit risk scoring. XGBoost and LightGBM models were compared, both of which resulted in 93% accuracy. My findings suggest LightGBM is marginally better for precision-focused lending, XGBoost offers superior sensitivity for identifying high-risk defaults. By integrating SHAP and LIME, I demonstrated that complex decision-tree models can be rendered transparent, providing both global feature importance and individual loan justifications.

Future work: my next project will be on fine-tuning of hyperparameters using Bayesian algorithms and the robustness testing of interpretability frameworks under adversarial attacks.

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